

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

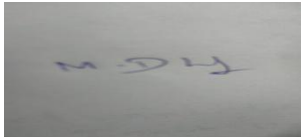
Total Marks: 30

Code of Conduct

I DHILIBAN M/192421166 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

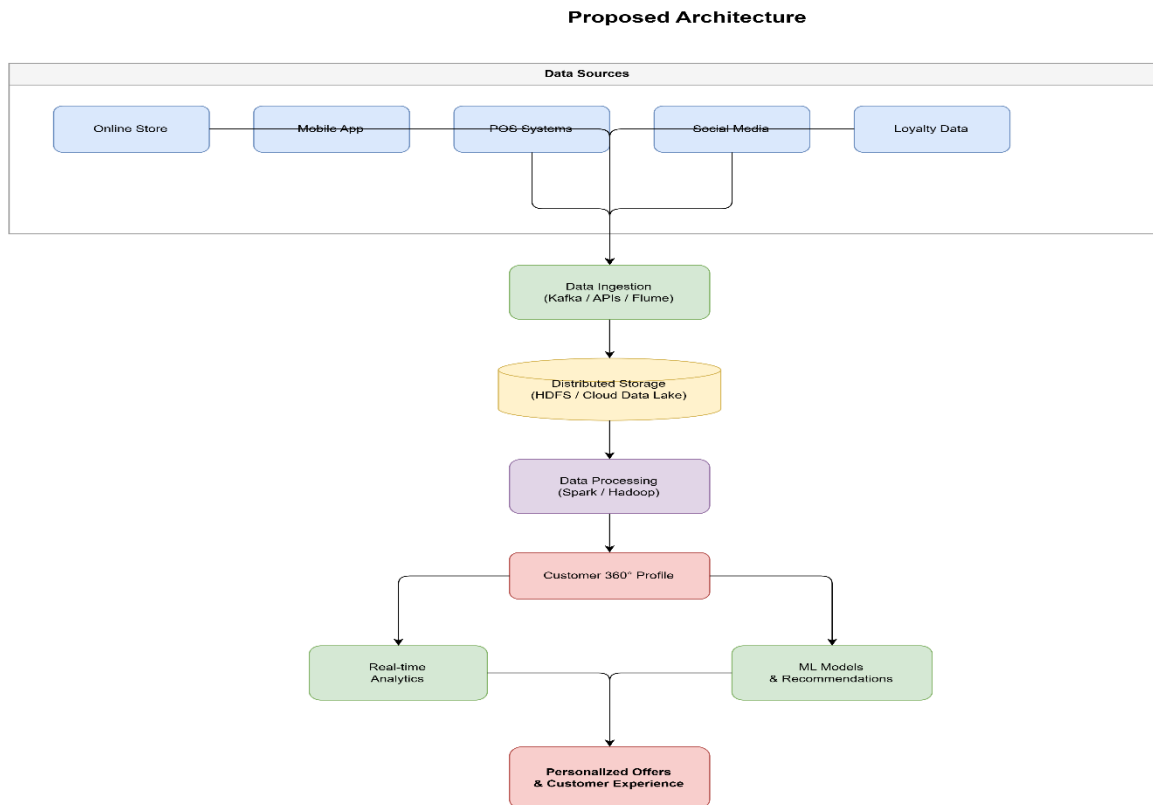
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Unified Big Data Architecture for a Multinational Retail Organization

A multinational retail organization may collect customer data from websites, mobile applications, physical stores, social media, loyalty programs, and payment systems. If this data is fragmented, it becomes difficult to understand customer behavior and provide personalized services.



Explanation:

Data from online and offline sources is collected using APIs and streaming technologies such as Apache Kafka. It is stored in a distributed data lake such as HDFS or cloud object storage. Apache Spark processes both batch and real-time data. A unified customer profile combines purchase history, browsing behavior, preferences, and location. Machine learning models then generate personalized product recommendations and offers.

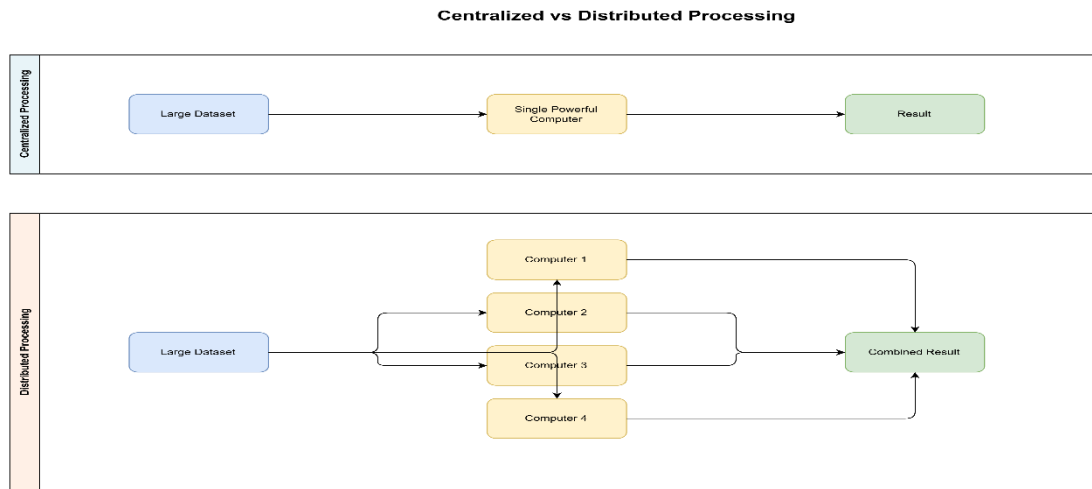
Example: If a customer searches for shoes online and previously purchased sportswear from a physical store, the system can immediately recommend suitable running shoes.

Benefits:

- 360° view of customers
- Real-time analytics
- Personalized recommendations
- Better customer retention
- Improved sales and marketing decisions

2. Distributed Computing for Large-Scale Scientific Research

Scientific research generates huge datasets from satellites, laboratories, simulations, medical research, and particle experiments. A centralized computer may take a very long time to process such data.



Distributed computing divides a large problem into smaller tasks and executes them simultaneously on multiple computers.

Technologies such as **Hadoop MapReduce** and **Apache Spark** support distributed processing.

Advantages

- **Parallelism:** Multiple tasks execute simultaneously.
- **Speed:** Processing time is significantly reduced.
- **Scalability:** More computers can be added when data increases.
- **Fault tolerance:** Failure of one node does not necessarily stop the entire system.
- **Cost efficiency:** Commodity computers can be combined instead of depending on one extremely powerful machine.

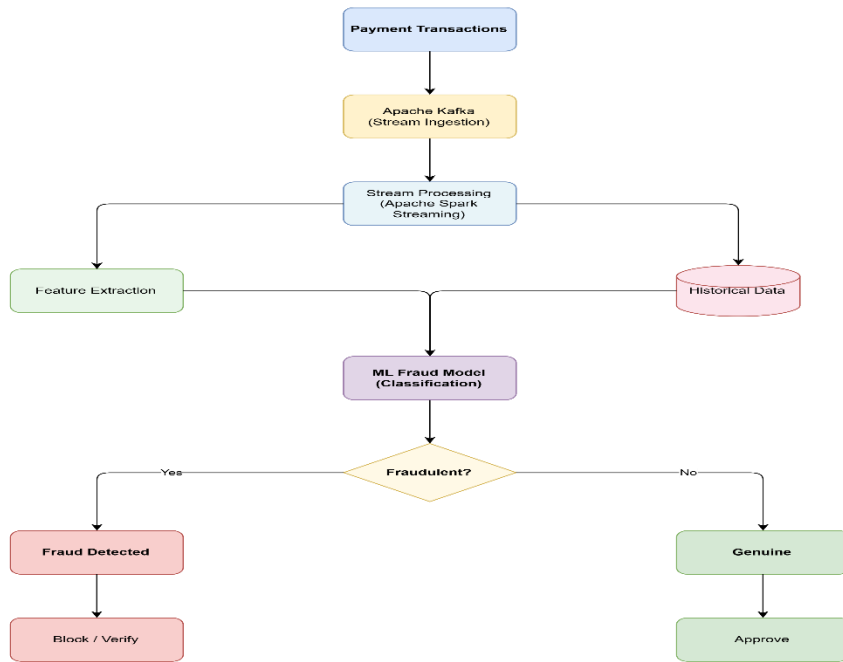
Example: In genomic research, millions of DNA sequences can be divided among multiple processing nodes and analyzed simultaneously.

Therefore, distributed computing provides better computational efficiency and scalability than a centralized architecture for large scientific datasets.

3. Fraud Detection Framework for a Digital Payment Platform

Digital payment systems generate continuous streams of transactions. Fraud detection must therefore analyze transactions in real time.

Payment Fraud Detection Framework



The system collects transaction amount, location, device information, transaction frequency, IP address, and user behavior. Kafka can ingest transactions continuously, while Spark Streaming processes them in real time.

Machine learning models such as Random Forest, Logistic Regression, or Gradient Boosting can calculate a fraud probability score.

Example: If a customer normally makes ₹500 transactions in Chennai but suddenly makes a ₹75,000 transaction from another country, the system can identify it as suspicious.

Benefits

- Real-time fraud detection
- Reduced financial losses
- Faster response to suspicious activity
- Detection of complex fraud patterns
- Improved prediction accuracy using historical data

Thus, combining streaming analytics with predictive models enables proactive rather than reactive fraud prevention.

4. Multimedia Data Management in Big Data

Multimedia data includes images, audio, and video. It creates several challenges because it is very large, unstructured, and requires high storage and processing capacity.

Major Challenges

1. **Huge volume:** High-resolution videos require large storage.
2. **Unstructured format:** Images, audio, and video do not follow simple tabular structures.
3. **High processing requirements:** Video analysis requires significant CPU/GPU resources.
4. **High bandwidth:** Transferring multimedia data requires high network capacity.
5. **Metadata management:** Information such as timestamp, location, format, and user must also be managed.
6. **Security:** Multimedia may contain sensitive personal information.

Recommended Technologies

Images —┐

Audio —┐→ Object Storage / HDFS

Video —┐ |



Spark Processing



AI/ML Analysis



Applications

Suitable technologies include:

- **HDFS:** Distributed storage for large datasets.
- **Cloud Object Storage:** Amazon S3, Azure Blob Storage, or Google Cloud Storage.
- **Apache Spark:** Large-scale multimedia processing.
- **NoSQL databases:** MongoDB or Cassandra for metadata.
- **Compression:** Reduces storage and network requirements.
- **CDNs:** Improve delivery of video and images.

A combination of distributed storage, compression, metadata databases, and parallel processing provides efficient multimedia management.

5. Disaster Recovery and Business Continuity for Cloud Big Data

A cloud-based big data platform must continue operating even when hardware, software, network, or data-center failures occur.

Key Strategies

- **Redundancy:** Maintain multiple servers and resources.
- **Replication:** Maintain copies of data in different locations.
- **Backup:** Periodically create recoverable copies.
- **Automated failover:** Automatically redirect users to a healthy system.
- **Multi-region deployment:** Protects against regional failures.
- **Monitoring:** Detects failures immediately.

Example: If the primary cloud region becomes unavailable, traffic can automatically move to a secondary region containing replicated data.

These mechanisms reduce downtime, prevent data loss, and maintain continuous business operations.

6. Impact of Data Quality Issues and Data Governance

Data quality directly affects the reliability of big data analytics. Poor-quality data can produce incorrect reports, inaccurate predictions, and bad business decisions.

Major Data Quality Problems

Problem	Impact
Inconsistency	Different values for the same information
Duplication	Same customer or transaction stored multiple times
Missing values	Incomplete analysis
Incorrect data	Wrong predictions and decisions
Outdated data	Poor real-time decision-making

Reliable Analytics

Organizations should establish:

- Data quality standards
- Data validation rules
- Master Data Management
- Deduplication procedures

- Missing-value handling
- Data ownership and stewardship
- Metadata management
- Access control and auditing

Example: If a customer appears three times with slightly different names, deduplication can combine the records into a single customer profile.

Good data governance improves trust, consistency, accuracy, and analytical reliability.

7. Scalable Recommendation System for an Online Streaming Service

A streaming service such as Netflix or a music platform generates huge amounts of user behavior data, including viewing history, search history, ratings, watch duration, skips, and likes.

The system analyzes user behavior and content characteristics. Distributed processing using Spark enables millions of user records to be processed efficiently.

Recommendation approaches include:

- Collaborative filtering
- Content-based filtering
- Hybrid recommendation
- Machine learning models

Example: If a user frequently watches action movies, the system can recommend newly released action movies with similar characteristics.

Benefits

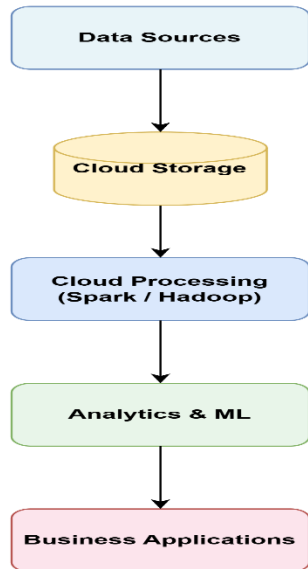
- Improved recommendation accuracy
- Personalized user experience
- Increased watch time
- Better customer engagement
- Scalable processing for millions of users

Thus, big data and distributed processing allow recommendations to be continuously updated according to user behavior.

8. Role of Cloud Computing in Big Data Ecosystems

Cloud computing provides on-demand computing, storage, networking, and analytics resources for big data applications.

Big Data Cloud Architecture



Advantages

- 1. Scalability:** Resources can be increased or decreased according to workload.
- 2. Cost Management:** Organizations can use pay-as-you-go services instead of purchasing large hardware infrastructure.
- 3. Flexibility:** Different storage, database, analytics, and machine learning services can be selected.
- 4. Faster Deployment:** Big data environments can be created quickly.
- 5. Global Availability:** Cloud services can support applications across different geographic regions.

Limitations

- Data security and privacy concerns
- Vendor lock-in
- Network dependency
- Unexpected cloud costs
- Compliance challenges
- Data transfer costs

Therefore, cloud computing offers excellent scalability and flexibility, but organizations must carefully manage security, cost, privacy, and vendor dependency.

9. Real-Time Analytics Framework for Transportation and Logistics

Transportation companies generate data from GPS devices, vehicles, traffic systems, fuel sensors, mobile applications, and delivery systems.

The system continuously receives GPS and traffic information. Stream processing analyzes current conditions while historical data helps predict traffic and delivery times.

Applications

- Route optimization
- Traffic prediction
- Fuel consumption monitoring
- Vehicle tracking
- Delivery time prediction
- Fleet management

Example: If a highway suddenly becomes congested, the system can identify an alternative route and send it to the driver.

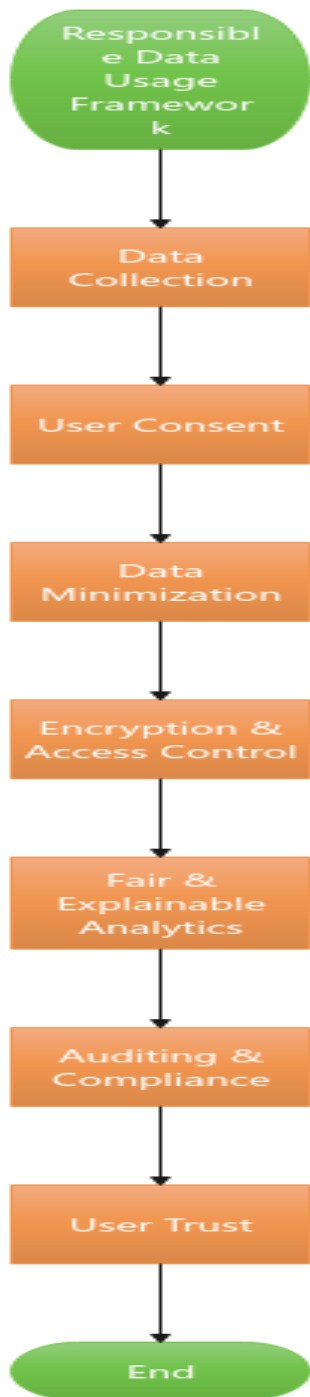
Big data analytics therefore reduces travel time and fuel consumption while improving delivery efficiency and operational decision-making.

10. Ethical and Legal Challenges of Big Data

Big data provides powerful analytical capabilities, but collecting and analyzing personal data can create ethical and legal problems.

Major Challenges

1. **Privacy:** Personal information may be collected without sufficient awareness.
2. **Consent:** Users should understand how their data is being used.
3. **Security:** Data breaches can expose sensitive information.
4. **Bias:** Biased datasets can produce discriminatory decisions.
5. **Transparency:** Users may not understand automated decisions.
6. **Data ownership:** Organizations must determine who can control and use collected data.
7. **Regulatory compliance:** Organizations must follow applicable data protection and privacy laws.



Recommended Policies

- Obtain informed user consent.
- Collect only necessary data.
- Encrypt sensitive information.
- Apply role-based access control.
- Anonymize or pseudonymize personal data where appropriate.

- Regularly audit algorithms for bias.
- Maintain clear data-retention and deletion policies.
- Provide transparency about automated decisions.
- Follow applicable privacy and data-protection regulations.
- Establish accountability through data governance teams.

Example: A company using customer location data for personalized advertising should clearly inform customers, obtain appropriate consent, protect the data, and avoid using it for unrelated purposes.

Responsible big data practices balance business benefits with privacy, fairness, security, and legal compliance. This improves long-term customer trust.